

Cost Analysis of AWS, GCP, and Azure for Kubernetes-Based MLOps Deployments

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Abstract—Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure are competing to capture the machine learning operations (MLOps) landscape. They offer an almost identical set of core MLOps components, including pipeline orchestration, model registries, and feature stores. As the functionality between providers is not much, the deciding factor for organizations often comes down to cost and the return on investment that each platform delivers. This survey examines the pricing models of aforementioned services for Kubernetes-based MLOps deployments. We compare the cost of control planes, worker nodes, object and block storage, networking and egress, load balancing, and observability. We then create a monthly cost prediction for a small-scale production deployment with 1 TB of stored data and 100 GB of monthly egress under minimum-redundancy assumptions, and we discuss challenges surrounding these services.

I. INTRODUCTION

Together, AWS, Azure, and GCP cover roughly 63% of the worldwide cloud infrastructure market [1]. Any organization building machine learning systems on the cloud needs to understand how these providers price the underlying components needed to take their solution to production. In this paper we focus mainly on Kubernetes-orchestrated deployments, as it provides high availability, self-healing, automated deployments, and efficient resource utilization, and reduces operational burden [2].

The pricing surfaces of these three providers are similar in most cases. However, small per-unit differences can compound quickly at the scale at which production ML systems are expected to operate. We aim to make those differences explicit and quantify them in a representative scenario. We only look at inference-time deployment costs, while training-related services and associated pricing are considered out of scope.

II. BACKGROUND

A. MLOps and Kubernetes

Machine learning operations (MLOps) are a set of practices that automate and simplify machine learning (ML) workflows and deployments. MLOps is an ML culture and practice that unifies ML application development (Dev) with ML system deployment and operations (Ops) [3]. It refers to the engineering discipline of operationalizing machine learning models, covering data ingestion, training, model registry, deployment, monitoring, and retraining. Modern MLOps stacks are increasingly run on managed Kubernetes (k8s) services because Kubernetes provides workload portability, declarative

resource management, and a rich ecosystem of operators for ML-specific concerns such as GPU scheduling and distributed training.

B. Managed Kubernetes Options

Each of the three providers offers a managed Kubernetes service: Amazon Elastic Kubernetes Service (EKS) on AWS [4], Google Kubernetes Engine (GKE) on GCP [5], and Azure Kubernetes Service (AKS) on Azure [6]. These services simplify cluster management by handling control plane operations, scaling, and maintenance, allowing users to focus primarily on building the solutions.

C. Why Cost Comparison is Not Simple

A comparison based only on the price of a control plane and worker nodes can be misleading. The total cost of ownership of a Kubernetes-based MLOps deployment is the sum of (i) control plane fees, (ii) worker-node machine, (iii) persistent block storage attached to nodes, (iv) object storage for datasets and model artifacts, (v) per-operation storage charges, (vi) load balancing, (vii) egress bandwidth, and (viii) logging and monitoring ingestion. A meaningful comparison must account for all of these line items.

III. DISCUSSION

A. Control Plane Pricing

The Kubernetes control plane is the first line item every cluster incurs. AWS EKS charges \$0.10 per cluster per hour, which works out to approximately \$73 per month per cluster [4]. GCP GKE also charges \$0.10 per cluster per hour but provides a \$74.40 monthly credit per billing account, which fully offsets the cost of one zonal cluster [5]. Azure AKS offers a Free tier with no control plane fee. But it comes without a service level agreement (SLA) backed uptime guarantee which may not be suitable for a production environment and a Standard tier at \$0.10 per cluster per hour for production workloads with an SLA [6].

For a single-cluster deployment, GCP is cheapest for control plane layer. For organizations running many clusters, GCP's per-account credit becomes proportionally less impactful.

B. Worker Node

We compare a representative 4 vCPU, 16 GB instance on each provider. These general-purpose instances can be

TABLE I
CONTROL PLANE COST COMPARISON (PER CLUSTER, MONTHLY)

Provider	List Price	Effective Cost
AWS EKS	\$0.10/hr	\$73.00
GCP GKE	\$0.10/hr	\$0.00 (with credit)
Azure AKS (Free)	\$0.00	\$0.00 (no SLA)
Azure AKS (Standard)	\$0.10/hr	\$73.00

considered a baseline for an ML inference or pipeline node and can be used to schedule k8s pods:

- **AWS m5a.xlarge**: \$0.172/hour [7]
- **GCP n4d-standard-4**: \$0.1694/hour [8]
- **Azure D4as_v5**: \$0.1820/hour [9]

At 730 hours per month, a single node costs approximately \$125.56 on AWS, \$132.86 on Azure, and \$123.66 on GCP. GCP holds a marginal lead at the on-demand price point for this generation of general-purpose VMs.

C. Storage

Object storage is where ML datasets, model artifacts, and pipeline outputs are persisted. The relevant unit prices are summarized in Table II.

TABLE II
OBJECT STORAGE PRICING (STANDARD / HOT TIER)

Item	AWS S3	Azure Blob	GCP GCS
Storage (per GB-month)	\$0.023	\$0.018	\$0.020
Write ops (per 10K)	\$0.050	\$0.065	\$0.050
Read ops (per 10K)	\$0.004	\$0.005	\$0.004

Azure Blob offers the lowest storage cost per GB per month [12], while AWS and GCP share the lowest pricing for read and write operations [10], [11]. But Azure charges a bit more on operations. For storage-heavy workloads (training data lakes), Azure is cheapest; for operation-heavy workloads, AWS and GCP are cheaper.

D. Block Storage for Worker Nodes

Block storage is attached to the worker nodes for the OS volume, scratch space, and PersistentVolumes. AWS EBS gp3 is priced at \$0.08 per GB-month [13], Azure Managed Disks at \$0.081 per GB-month [15], and GCP Persistent Disk at approximately \$0.17 per GB-month for SSD-backed disks [14]. AWS and Azure are roughly equivalent and substantially cheaper than GCP at the SSD persistent-disk tier.

E. Networking: Load Balancers and Egress

MLOps cluster should front its inference services with a managed load balancer. AWS Application Load Balancer is \$0.0225 per ALB-hour [16], Azure Standard Load Balancer is \$0.025 per hour [18], and GCP Cloud Load Balancing is \$0.025 per forwarding-rule-hour [17]. Over a 730-hour month this is \$16.43, \$18.25, and \$18.25 respectively.

Egress, the bandwidth leaving the cloud provider is often the most underestimated cost in cloud and an unforeseen cost

that companies incur. AWS charges \$0.09 per GB [7], Azure charges \$0.087 per GB [20], and GCP charges \$0.085 per GB [19] for general internet egress after their free tier. AWS and Azure provide first 100 GB free while GCP provides a 200 GB free tier. Egress can dominate cost for ML services that return large prediction payloads to external clients.

F. Monitoring and Logging

Observability ingestion is the most asymmetric line item across the three providers, as shown in Table III.

TABLE III
LOGGING INGESTION PRICING [25]–[27]

Service	Free Tier	Per-GB Ingest
AWS CloudWatch Logs	5 GB/month	\$0.50
GCP Cloud Logging	50 GB/month	\$0.50
Azure Monitor	5 GB/workspace/month	\$2.76

GCP is significantly more cost-effective for high-frequency logging workloads due to its 50 GB free tier. Azure is more than $5\times$ more expensive per GB ingested than the other two and also has a comparatively lower free tier. For ML systems that have a lot of request logs, prediction traces, or metrics, this item can become the largest non-compute cost on Azure.

G. Cluster Scalability Ceilings

The maximum supported nodes per cluster differ by orders of magnitude:

- **AWS EKS**: up to 100,000 worker nodes per cluster [21]
- **GCP GKE**: 65,000 nodes per cluster [22], [24]
- **Azure AKS**: 5,000 nodes per cluster [23]

For organizations whose roadmap includes very large training fleets, this ceiling is a real architectural constraint. Azure’s 5,000-node ceiling forces a federated, multi-cluster design at scale, which can result in separate operational and cost overhead.

IV. COST PREDICTION SCENARIO

We model a small company trying to go to market with their ML solution. We assume running a single Kubernetes-based MLOps deployment is used with the following assumptions, chosen to reflect minimum redundancy rather than zero redundancy:

- One managed Kubernetes cluster
- Two worker nodes (4 vCPU, 16 GB each) for high availability
- One managed load balancer
- 1 TB of object storage (training data, artifacts)
- 200 GB of block storage across both nodes
- 100 GB of monthly egress (web responses)
- Approximately 100,000 write and 1,000,000 read object-storage operations
- Approximately 10 GB/month of ingested logs
- 730 hours of runtime, on-demand US-region pricing

The resulting cost breakdown is shown in Table IV

TABLE IV
ESTIMATED MONTHLY COST BREAKDOWN FOR THE REFERENCE SCENARIO (USD)

Line Item	AWS	GCP	Azure
Kubernetes control plane	73.00	0.00 (after \$74.40 credit)	73.00
Worker nodes (2×4 vCPU/16 GB, 730 h)	251.12	247.32	265.72
Block storage (200 GB)	16.00	34.00	16.20
Object storage (1 TB)	23.55	20.48	18.43
Object-storage write operations (100K)	0.50	0.50	0.65
Object-storage read operations (1M)	0.40	0.40	0.50
Load balancer (730 h)	16.43	18.25	18.25
Egress (100 GB)	0	0	0
Logging ingestion (10 GB, after free tier)	2.50	0.00 (within 50 GB free)	13.80
Estimated Monthly Total	383.50	320.95	406.55

Several observations follow from Table IV. First, the spread between the cheapest and most expensive provider in this scenario is approximately \$85 per month, or about 26% of the lowest total. Second, GCP wins overall, primarily because the GKE control plane credit absorbs the \$73 fee which is the approximate difference between highest and lowest. Third, GCP’s higher block-storage cost partially offset its other gains. If the workload were block-storage-heavy or required very large attached SSDs, the ranking would change. Fourth, Azure becomes substantially expensive once an SLA is required (rather than free AKS tier) and exceeds AWS on total cost.

The dominant cost driver in all three columns is worker-node (between 65% and 77% of the bill), which means the most consequential cost-engineering is using the machine of right-size rather than not switching providers.

V. CHALLENGES

A. Hidden and Asymmetric Costs

Egress is the most consistently underestimated line item in cloud cost modeling. A workload that is cheap to run inside a single region can become prohibitively expensive once predictions, embeddings, or feature payloads must leave the provider’s network. AWS’s egress rate of \$0.09/GB can be particularly punishing for prediction APIs that serve large external client populations.

Logging is similarly asymmetric. A team that designs verbose tracing and logging on AWS or GCP and later migrates to Azure will see its observability bill multiply because Azure Monitor charges \$2.76/GB versus \$0.50/GB on the other two platforms.

B. Scalability Ceilings as Architectural Risk

The gap between Azure’s 5,000-node-per-cluster ceiling and AWS’s 100,000-node ceiling can be a big concern for organizations training foundation models or running very large feature stores. Hitting the ceiling forces a redesign around cluster management, cross-cluster service meshes, and multi-cluster CI/CD. All of that will add real operational and monetary cost, which may not be forecasted directly.

C. Vendor Lock-In Despite Kubernetes

Although Kubernetes is portable in principle, the surrounding services (managed databases, object storage with provider-specific IAM, identity, secret management, and observability backends) are not. An MLOps stack that binds tightly to S3 IAM roles, CloudWatch Logs Insights queries, and SageMaker endpoints is portable in name only. The same caveat applies symmetrically to GCS, Cloud Logging, and Vertex AI on GCP, and to Blob Storage, Log Analytics, and Azure ML on Azure.

D. On-Demand Pricing as a Pessimistic Baseline

The figures in this paper are all on-demand prices. Real production deployments use a mix of reserved instances, committed-use discounts, savings plans, and spot nodes. These mechanisms can reduce cloud expense, but they can also bind the customer to multi-year commitments and to specific instance families. This reduces the flexibility teams try to gain by using k8s orchestration in the first place.

VI. CONCLUSION

At the small-scale, single-cluster, minimum-redundancy operating point modeled in this paper, GCP is the cheapest of the three providers (approximately \$321/month), followed by AWS and Azure, although it can be argued that the difference is not that big. Besides, each platform has some components that are less costly than others. Companies often have to adapt according to the platform they are choosing or go with the platform that offers them the best value, which can be both cost and flexibility.

The estimated ranking is not stable across workloads. Log-intensive workloads tend to penalize Azure more significantly. AWS generally provides strong connectivity within its own components but lacks in flexibility. Google Cloud is often the first to introduce new innovations, but it may lag in ecosystem maturity and integration capabilities.

Therefore, a robust cloud-selection decision for MLOps should not be based on a single monthly total. It should be based on the necessities of the actual workload, the solution requirements, the realistic scalability expectations, reliability and availability that the organization is willing to fund.

REFERENCES

- [1] Statista, “Worldwide Market Share of Leading Cloud Infrastructure Service Providers.” [Online]. Available: <https://www.statista.com/chart/18819/worldwide-market-share-of-leading-cloud-infrastructure-service-providers/>
- [2] Kubernetes Documentation, “Overview of Kubernetes.” [Online]. Available: <https://kubernetes.io/docs/concepts/overview>
- [3] Amazon Web Services (AWS), “What is MLOps?.” [Online]. Available: <https://aws.amazon.com/what-is/mlops/>
- [4] Amazon Web Services, “Amazon EKS Pricing.” [Online]. Available: <https://aws.amazon.com/eks/pricing/>
- [5] Google Cloud, “Google Kubernetes Engine Pricing.” [Online]. Available: <https://cloud.google.com/kubernetes-engine/pricing>
- [6] Microsoft Azure, “Azure Kubernetes Service (AKS) Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/kubernetes-service/>
- [7] Amazon Web Services, “Amazon EC2 On-Demand Pricing.” [Online]. Available: <https://aws.amazon.com/ec2/pricing/on-demand/>
- [8] Google Cloud, “General-Purpose VM Pricing.” [Online]. Available: <https://cloud.google.com/products/compute/pricing/general-purpose>
- [9] Microsoft Azure, “Linux Virtual Machines Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/virtual-machines/linux/>
- [10] Amazon Web Services, “Amazon S3 Pricing.” [Online]. Available: <https://aws.amazon.com/s3/pricing/>
- [11] Google Cloud, “Cloud Storage Pricing – Operations by Class.” [Online]. Available: <https://cloud.google.com/storage/pricing#operations-by-class>
- [12] Microsoft Azure, “Azure Blob Storage Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/storage/blobs/>
- [13] Amazon Web Services, “Amazon EBS Pricing.” [Online]. Available: <https://aws.amazon.com/ebs/pricing/>
- [14] Google Cloud, “Disks and Image Pricing.” [Online]. Available: <https://cloud.google.com/compute/disks-image-pricing#disk>
- [15] Microsoft Azure, “Managed Disks Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/managed-disks/>
- [16] Amazon Web Services, “Elastic Load Balancing Pricing.” [Online]. Available: <https://aws.amazon.com/elasticloadbalancing/pricing/>
- [17] Google Cloud, “Cloud Load Balancing Pricing.” [Online]. Available: <https://cloud.google.com/load-balancing/pricing>
- [18] Microsoft Azure, “Load Balancer Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/load-balancer/#pricing>
- [19] Google Cloud, “Network Service Tiers pricing” [Online]. Available: <https://cloud.google.com/network-tiers/pricing>
- [20] Microsoft Azure, “Bandwidth Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/bandwidth/>
- [21] Amazon Web Services, “Amazon EKS now supports 100,000 worker nodes per cluster,” AWS What’s New, 2025. [Online]. Available: <https://aws.amazon.com/about-aws/whats-new/2025/07/amazon-eks-100000-worker-nodes-per-cluster/>
- [22] Google Cloud, “GKE Quotas and Limits.” [Online]. Available: <https://docs.cloud.google.com/kubernetes-engine/quotas>
- [23] Microsoft Learn, “Best practices for performance and scaling for large workloads in AKS.” [Online]. Available: <https://learn.microsoft.com/en-us/azure/aks/best-practices-performance-scale-large>
- [24] Google Cloud, “Planning for Large GKE Clusters.” [Online]. Available: <https://docs.cloud.google.com/kubernetes-engine/docs/concepts/planning-large-clusters>
- [25] Microsoft Azure, “Azure Monitor Pricing.” [Online]. Available: <https://azure.microsoft.com/en-us/pricing/details/monitor/>
- [26] Google Cloud, “Observability (Cloud Logging & Monitoring) Pricing.” [Online]. Available: <https://cloud.google.com/products/observability/pricing>
- [27] Amazon Web Services, “Amazon CloudWatch Pricing.” [Online]. Available: <https://aws.amazon.com/cloudwatch/pricing/>